

Monte Carlo Simulation Uge 2

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1 The Inverse Transform Method

We discussed the inverse transform method. This method is useful for generating random variables from a given distribution. However, we are restricted to distributions with a known cumulative distribution function (CDF) that is invertible. The method is based on the fact that if U is a uniform random variable on the interval $[0, 1]$, then $X = F^{-1}(U)$ has the distribution function F . Practically then, we can simulate a random variable X with distribution function F by generating a uniform random variable U and then computing $X = F^{-1}(U)$.

2 Discrete Distributions

Sampling from is a bit different and requires algorithms that act on the uniform random variable U to generate a discrete random variable X . To read more about this, refer to section 2.2 in the lecture notes.

3 Location-Scale Families

We say that a family of distributions is a location-scale family if a random variable X with distribution $f(x)$ can be transformed into a random variable $Z = \mu + \sigma X$, such that $f_Z(z) = f(x; \mu, \sigma) := \frac{1}{\sigma} f(\frac{z-\mu}{\sigma})$ also is a well-defined distribution.